

PROJECT REPORT

Title

CertifyX – AI-Powered Digital Certificate Management and Verification System Hackathon Project Report

1. Abstract

Educational institutions continue to rely on paper-based certificates and manual verification processes, which often result in delays, document forgery, duplicate certificates, and administrative workload. Verification of certificates generally requires physical inspection or manual confirmation from the issuing authority, making the process slow and inefficient.

CertifyX is a web-based digital certificate management platform that simplifies certificate generation, issuance, storage, and verification. The system enables institutions to issue digitally signed certificates with unique Certificate IDs and QR codes. Students can request and download certificates online, while staff members can approve, generate, and manage certificates through a dedicated dashboard. Every issued certificate can be verified instantly using its unique certificate ID or QR code.

The platform is built using **React, TypeScript, Express.js, Node.js, Vite**, and is deployed on **Render**, making it accessible from anywhere.

2. Problem Statement

Educational institutions face several challenges in certificate management:

- Manual certificate creation consumes significant administrative time.
- Verification requires physical interaction or email communication.
- Fake certificates are difficult to detect.
- Duplicate certificates can be created without proper tracking.
- Students must frequently visit the institution for simple documents.
- Paper-based certificates can be lost or damaged.
- Institutions lack a centralized digital certificate repository.

These issues reduce efficiency and increase administrative workload.

3. Proposed Solution

CertifyX provides a centralized digital certificate management system with separate portals for students and staff.

The system allows students to request certificates online while enabling staff members to approve, generate, digitally issue, and verify certificates. Every generated certificate contains:

- Unique Certificate ID
- QR Code
- Digital Signature
- Institution Details
- Student Information

Certificates can be verified instantly using the verification portal.

4. Objectives

The objectives of CertifyX are:

- Digitize certificate generation.
 - Reduce manual administrative work.
 - Prevent certificate forgery.
 - Enable instant certificate verification.
 - Maintain centralized certificate records.
 - Improve accessibility for students.
 - Ensure authenticity using QR-based verification.
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5. Existing System

Current systems generally involve:

- Manual form submission
- Physical approval
- Printed certificates
- Manual verification
- Long waiting periods

Limitations

- Time consuming
 - Paper dependent
 - High administrative effort
 - Easy duplication
 - Difficult verification
 - No centralized database
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6. Proposed System

The proposed system consists of two user roles.

Student

Students can:

- Login securely
 - Request certificates
 - Generate selected certificates
 - View issued certificates
 - Download certificates
 - Verify certificates
 - Track request status
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Staff

Staff members can:

- Login securely
- View student requests
- Approve or reject requests
- Generate certificates
- Issue certificates
- Maintain certificate records
- Verify certificate authenticity

7. Key Features

Student Dashboard

- Secure Login
- Certificate Requests
- Certificate Download
- Certificate Verification
- QR Code Access
- View Issued Certificates

Staff Dashboard

- Staff Login
- Student Request Management
- Certificate Generation
- Digital Signature
- QR Code Generation
- Certificate Database Management

Verification Module

- Enter Certificate ID
- Scan QR Code
- Display Certificate Status
- Show Certificate Details
- Detect Invalid Certificates

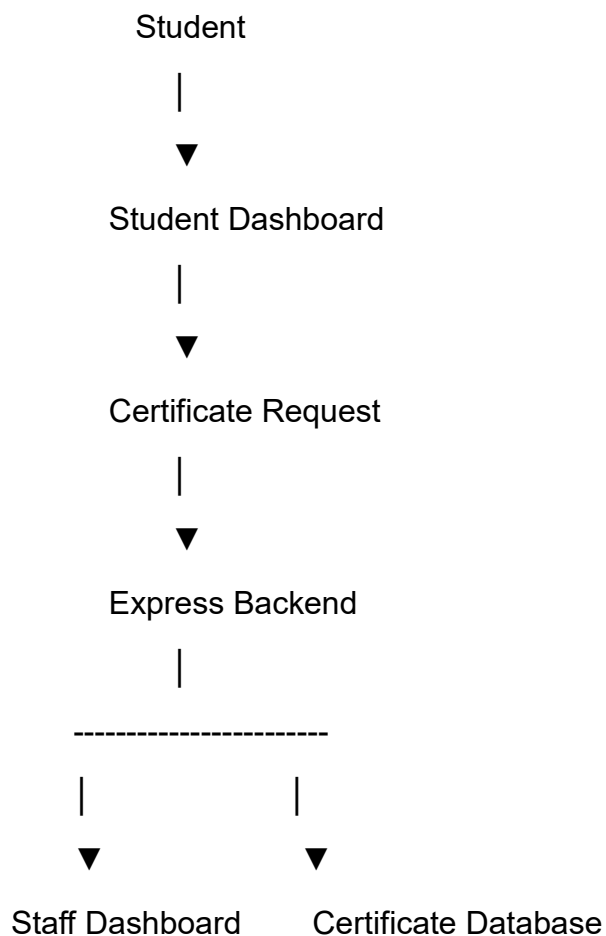
8. Technology Stack

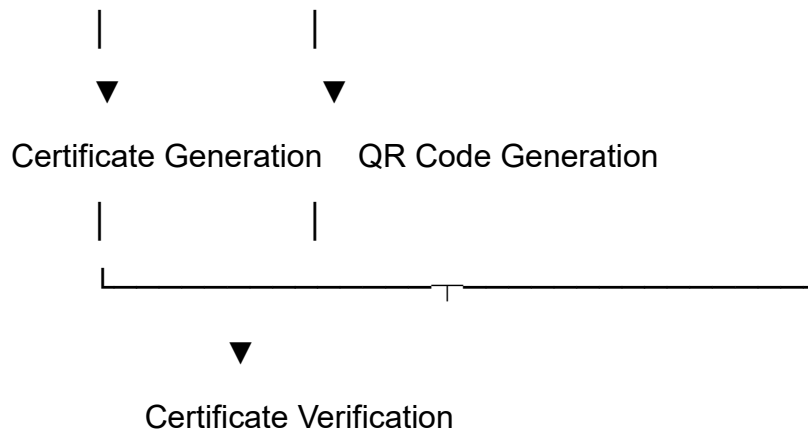
Component	Technology
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Frontend	React.js
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Component	Technology
Language	TypeScript
Backend	Express.js
Runtime	Node.js
Build Tool	Vite
Styling	Tailwind CSS
QR Generation	QRCode Library
OCR	Tesseract.js
Deployment	Render
Version Control	GitHub
IDE	Visual Studio Code

9. System Architecture





10. Modules

Module 1

Student Authentication

- Student Login
- Session Handling

Module 2

Staff Authentication

- Staff Login
- Dashboard Access

Module 3

Certificate Generation

- Select Certificate Type
- Enter Required Details
- Generate Certificate
- Generate Unique Certificate ID
- QR Code Generation
- Digital Signature

Module 4

Certificate Repository

- Store Certificates
 - Retrieve Certificates
 - View Issued Certificates
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Module 5

Certificate Verification

- Search using Certificate ID
 - QR Code Verification
 - Authenticity Validation
 - Invalid Certificate Detection
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11. Workflow

Student Login

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Request Certificate

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Staff Reviews Request

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Certificate Generated

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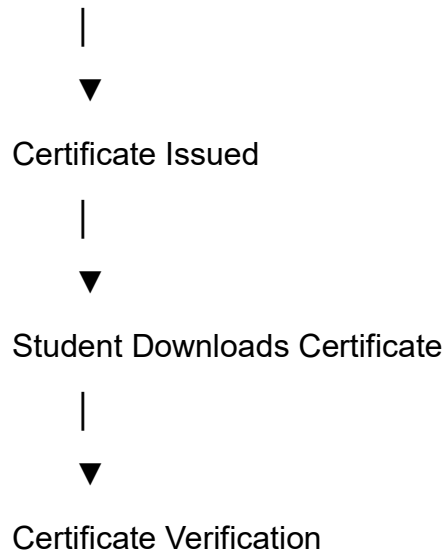


Unique Certificate ID Created

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QR Code Generated



12. Novelty of the Project

Unlike conventional certificate systems, CertifyX offers:

- Digital certificate generation
 - QR-based verification
 - Unique Certificate IDs
 - Online student requests
 - Staff approval workflow
 - Centralized certificate management
 - Reduced paperwork
 - Faster verification
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13. Advantages

- Paperless workflow
- Faster certificate generation
- Secure verification
- Easy accessibility
- Reduced administrative effort
- Prevents forgery
- Centralized management

- User-friendly interface
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14. Challenges Faced

During development:

- Deploying the application on Render.
 - Exporting the project from Lovable.
 - Configuring GitHub repository.
 - Connecting deployment with GitHub.
 - Managing frontend and backend together.
 - Debugging certificate verification.
 - Fixing deployment and build issues.
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15. Results

The developed application successfully:

- Provides separate Student and Staff dashboards.
 - Generates digital certificates.
 - Generates unique Certificate IDs.
 - Creates QR-enabled certificates.
 - Enables online certificate verification.
 - Runs as a live web application.
 - Supports centralized certificate management.
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16. Future Enhancements

Future improvements include:

- Blockchain-based certificate storage.
- AI-powered fake certificate detection.
- OCR verification from uploaded certificates.
- Email and SMS notifications.
- Multi-college support.

- Mobile application.
 - Admin analytics dashboard.
 - Role-based permissions.
 - Cloud database integration.
 - Automated approval workflows.
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17. Conclusion

CertifyX modernizes the traditional certificate management process by providing a secure, paperless, and efficient digital platform. It minimizes administrative workload while improving accessibility for students and ensuring certificate authenticity through unique certificate IDs and QR-based verification.

The system demonstrates how modern web technologies can simplify institutional workflows, reduce document fraud, and provide a scalable solution for educational organizations. With future enhancements such as blockchain verification, AI-assisted fraud detection, and cloud integration, CertifyX has the potential to evolve into a comprehensive digital credential management platform suitable for schools, colleges, universities, and training institutions.